



SARDAR PATEL UNIVERSITY

BCA(Semester- 4)(OLD)(NC) Examination April- 2025

US04SBCA55: SYSTEM SECURITY

Date: 05/04/2025 (Saturday) Time: 02:00 P.M To 04:00 P.M

Total: 50Marks

Q-1 Multiple Choice Questions

[10]

1. _____ Attacks are in the nature of eavesdropping on, or monitoring of transmissions.
a).Passive b).Active c).Threat d).None of these
2. The _____ service is concerned with assuring that a communication is Authentic.
a). Authentication b). Access control c). Confidentiality d).Integrity.
3. Traffic analysis is the type of _____ attack.
a). Passive attack b).Active Attack c). Service attack d).None of these
4. _____ is the scrambled message produced as output.
a). Encryption algorithm b).Plaintext c). Secret key d). Cipher text
5. A _____ cipher reorders symbols in a block of symbols.
a). Block b). Stream c). Substitution d). Transposition
6. Public-key cryptography is also Known as _____ cryptography.
a). Asymmetric-key b). symmetric-key c).Secret-key d).Private-key
7. _____ is an individual who is not authorized to use the computer.
a).Masquerader b).Misfeasor c). Clandestine user d). None of these
8. _____ virus Infects files with macro code that is interpreted by an application.
a). E-mail b). Macro c).Encrypted d).Polymorphic
9. A _____ firewall filters at the application layer.
a). Proxy b).Circuit-level c). Packet-filter d).Host-Based
10. A Packet filter firewall filters at the network or _____ layer.
a).Physical b).Application c). Data link d). Transport

Q-2 Short Answer Question (Attempt EIGHT out of TEN)

[16]

- 1 What is Threat & Attack? Explain in short.
- 2 Explain DATA INTEGRITY Security Service in short.
- 3 What is Security Service? Explain in short.
- 4 What is Cryptanalysis? Explain in short.
- 5 Define Plaintext and Cipher text in short
- 6 Define Encryption and Decryption algorithm in short
- 7 Explain E-mail Viruses in short.
- 8 Explain any two Antivirus Approaches in short.
- 9 What is Digital Signature? Explain in short.
- 10 Explain VPN (Virtual Private Network) Network in short.

Q-3

- (A) Explain Security Attacks in Detail.. [03]
 - (B) Explain Model for Network Security in Detail. [03]
- OR
- (A) Explain Computer Security Concept in Detail. [03]
 - (B) Explain Security Services in Detail. [03]

C.P.T.O.)

Q-4

(A) Explain Introduction to Cryptography in detail. [03]

(B) Explain Substitution & Transposition ciphers in detail. [03]

OR

(A) Explain types of attack on Encrypted messages in detail. [03]

(B) Explain Public key (Asymmetric-key) Cryptography in detail. [03]

Q-5

(A) Explain Intruders in detail [03]

(B) Explain Backdoor (Trapdoor) and Logic Bomb in detail. [03]

OR

(A) Explain Classification (Types) of viruses in detail. [03]

(B) Explain Antivirus approaches in detail. [03]

Q-6

(A) Explain Design Principles Firewalls. [03]

(B) Explain goal, scope, limitations, and techniques of Firewall in detail. [03]

OR

(A) Explain Digital Signature in detail. [03]

(B) Explain Firewalls Location and Configuration in detail.. [03]

